UNIMORE
UNIVERSITÀ DEGLI STUDI DI MODENA E REGGIO EMILIA

Presented by :
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Evaluating Frozen Foundation Model Embeddings for Morphological Correlation and Survival Analysis

Problem Statement

PROBLEM

Different foundation models can generate different representations from histopathological images, but their effectiveness for downstream computational pathology tasks needs to be evaluated. This project investigates and compares the performance of the UNI and GIGA models using multiple datasets and experimental approaches.

Objectives

01 Features Extraction

02 Ground Truth Generation

03 Clustering

04 Clustering Evaluation

05 Ground Truth vs Clusters

06 MIL Based Analysis

07 MIL Evaluation

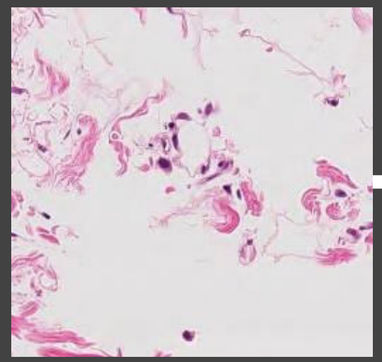
08 Results & Comparison

09 Conclusion

Features

Goal:

Extract meaningful feature representations from histopathological data using UNI and GIGA.

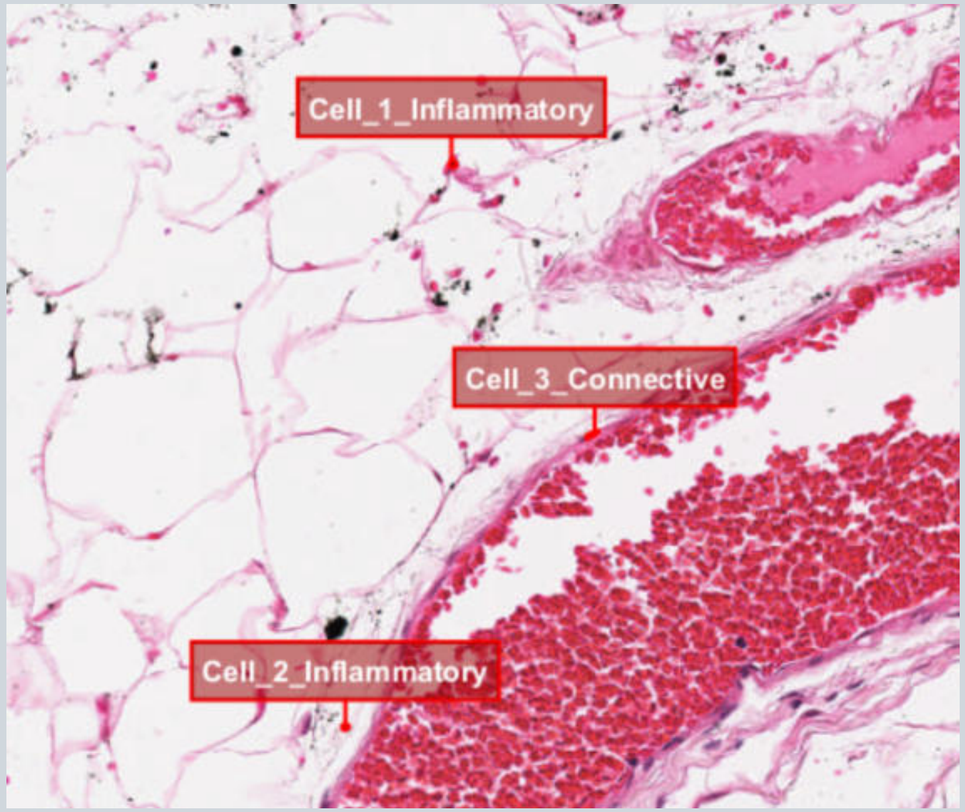


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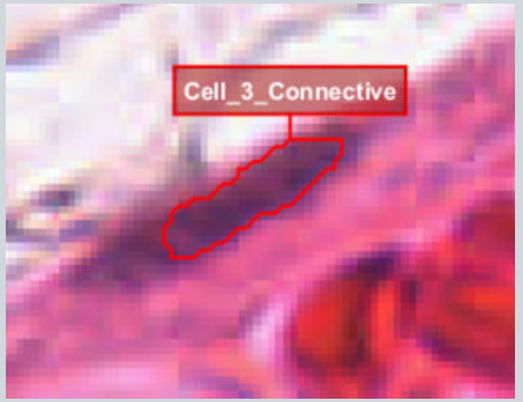
Ground Truth Generation

Pipeline

- Whole Slide Image (WSI)
- CellViT
- Cell Detection & Classification
- Cell Type + Cell Coordinates
- Ground Truth Data


Cell_1_Inflammatory
Cell_3_Connective
Cell_2_Inflammatory


Cell_1_Inflammatory


Cell_3_Connective


Cell_2_Inflammatory

Clustering

01 Algorithm

K-Means clustering to group similar cell features.

02 Number of Clusters

K=5, representing five cell groups.

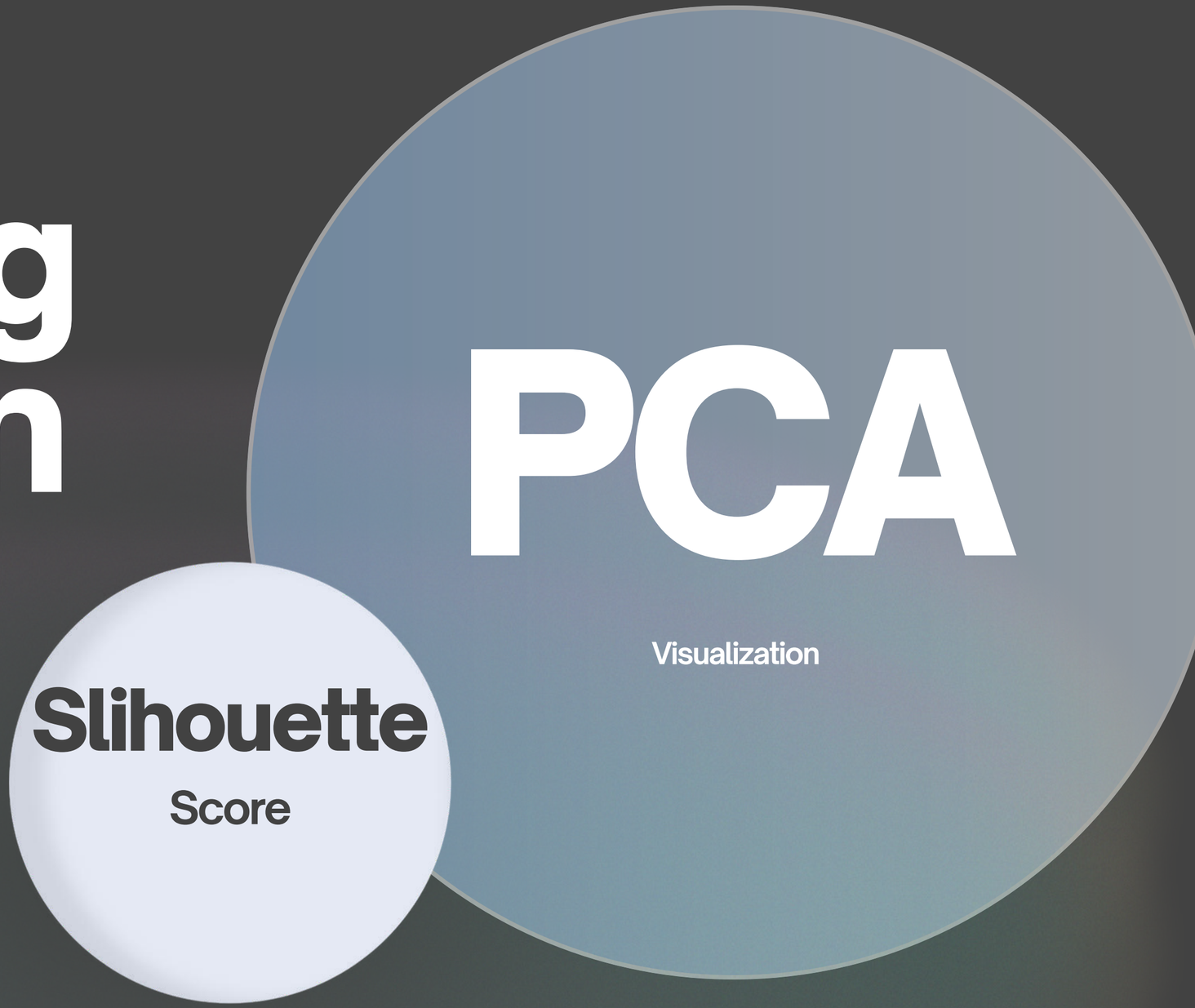
03 Input

Embeddings extracted using UNI and GIGA

04 Reference

Five cell types identified by CellViT

Clustering Evaluation



Ground Truth

Clusters

vs

Goal:

- Determine the cell-type composition of each K-Means cluster using the CellViT ground truth.

Method

- Assign each patch to one of the 5 K-Means clusters.
- Collect all patches belonging to each cluster.
- Retrieve the CellViT cell types from those patches.
- Count the number of cells of each type within each cluster.
- Analyze the dominant cell types in each cluster.

MIL Analysis



DataSets

- BRCE
- KIRP
- KIRC
- LUAD
- LUSC

Features

- UNI
- GIGA

MIL Methods

- ABMIL
- CLAM

MIL Evaluation

- Kaplan-Meier Survival Curves
- **C-Index**
- Cox Loss



ABMIL Couve for UNI and GIGA

Results & Comparison

PCA

Silhoutte score

Cell Vit

ABMIL KSMC

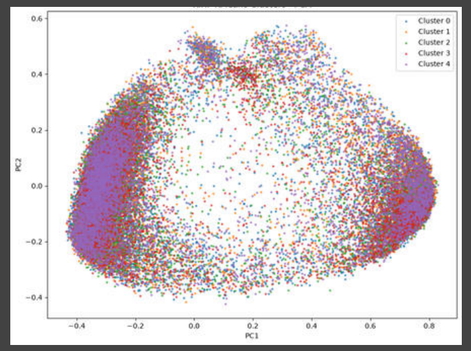
CLAM KSMC

Cox Loss

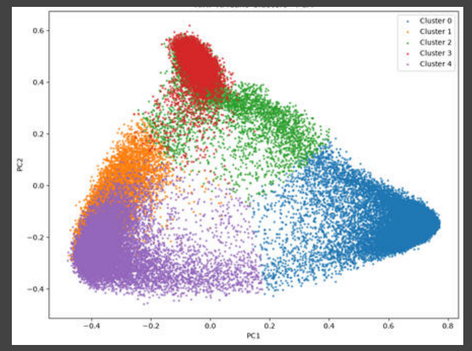
C-Index

PCA

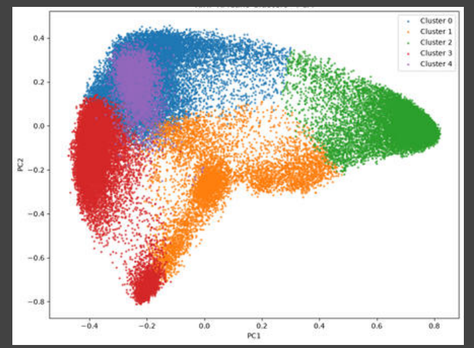
GIGA



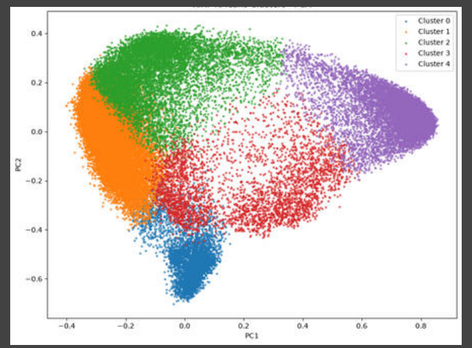
LUAD



LUSC

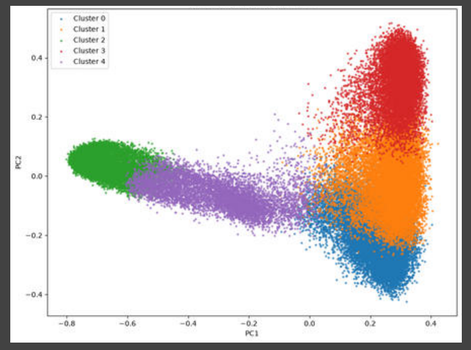


KIRP

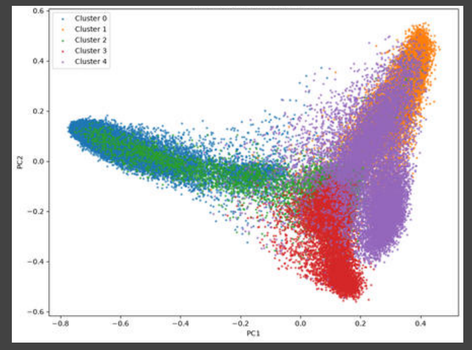


KIRC

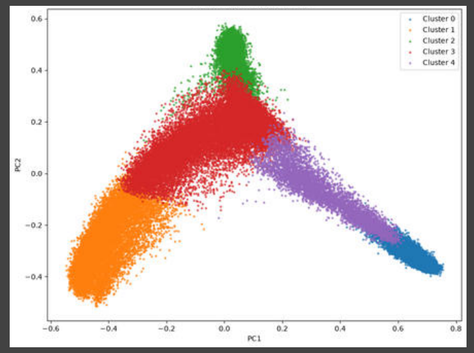
UNI



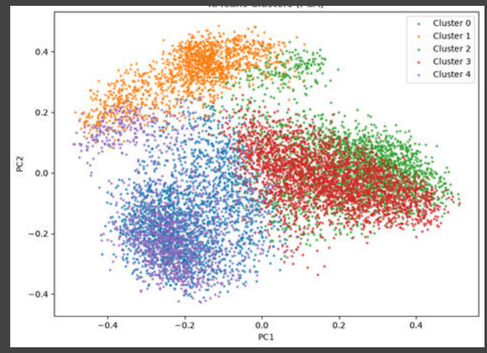
LUAD



LUSC

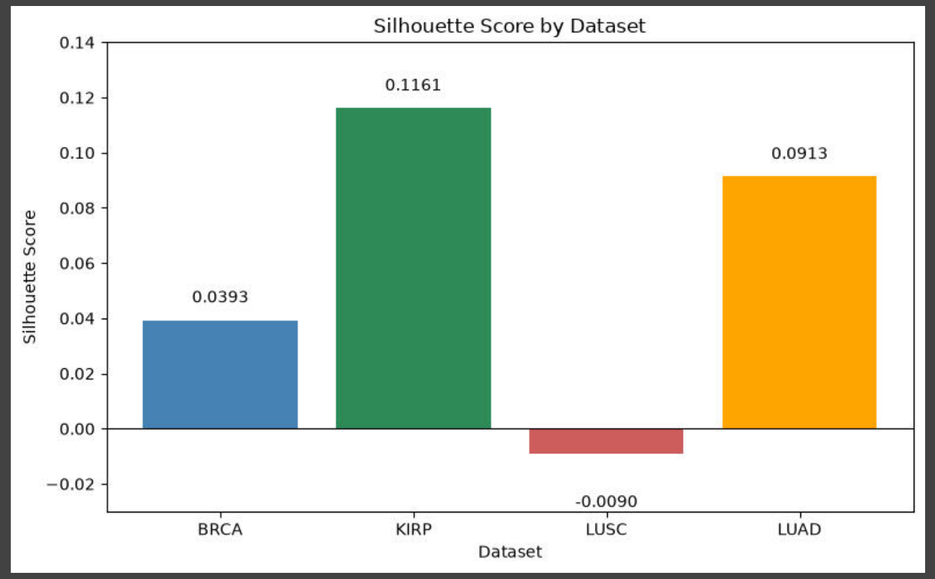


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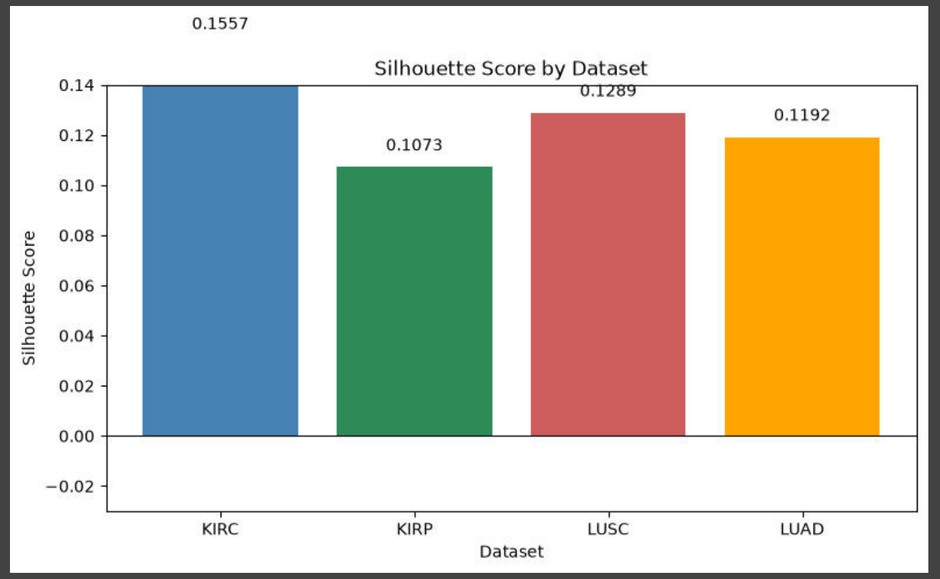


BRCA

Silhoutte Score



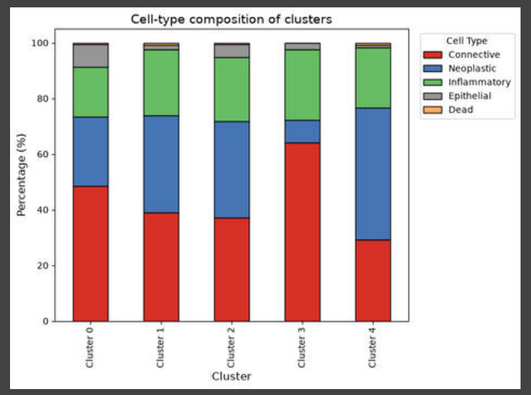
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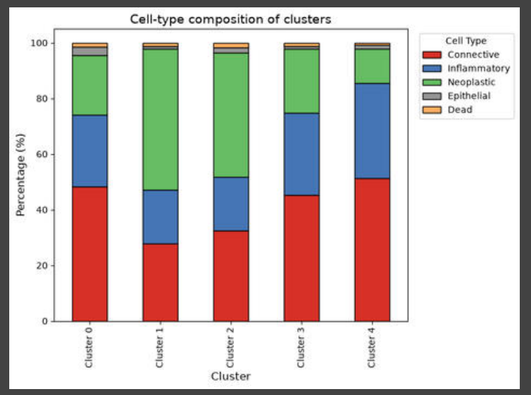
GIGA

Cell Vit

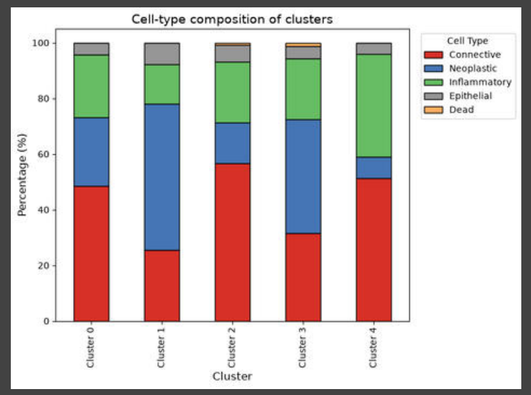
GIGA



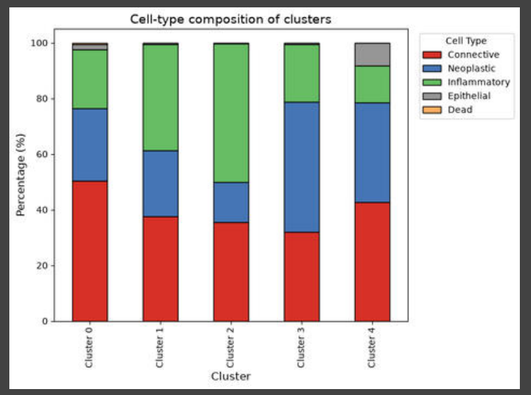
LUAD



LUSC

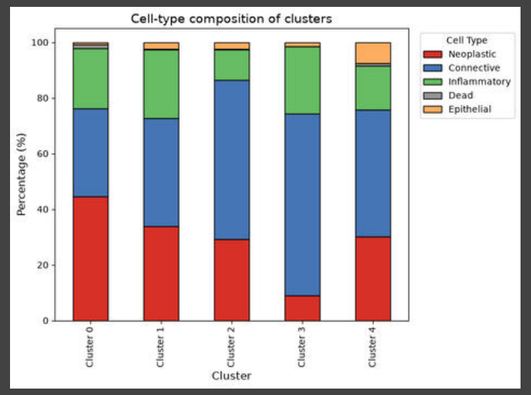


KIRP

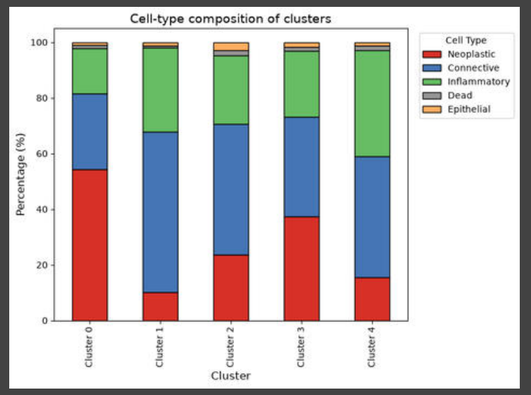


KIRC

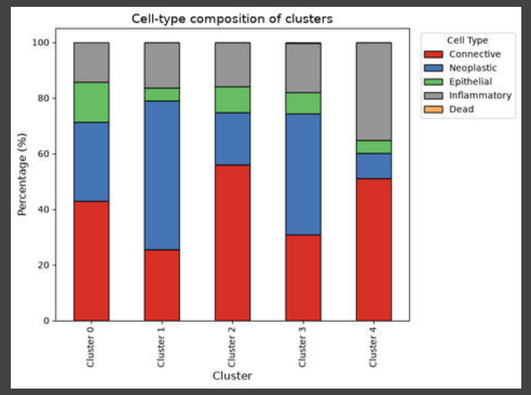
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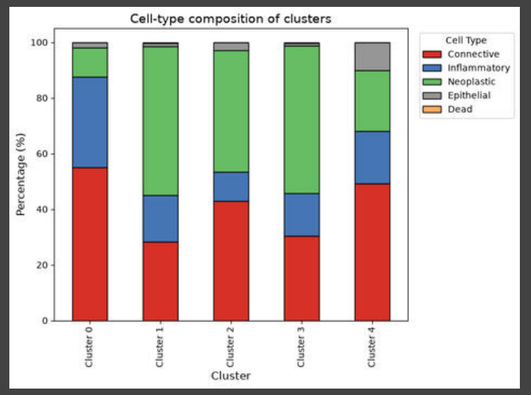
LUAD



LUSC



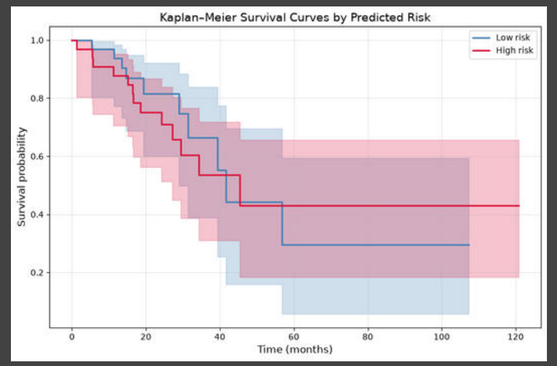
KIRP



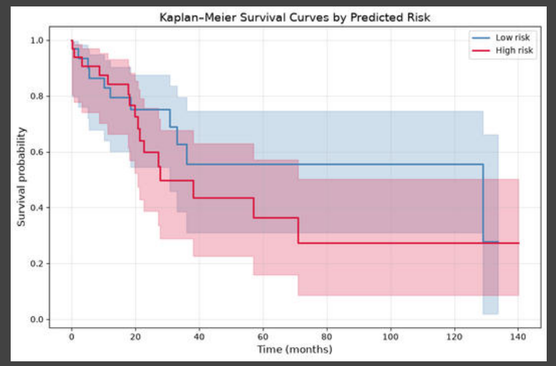
BRCA

ABMIL KSMC

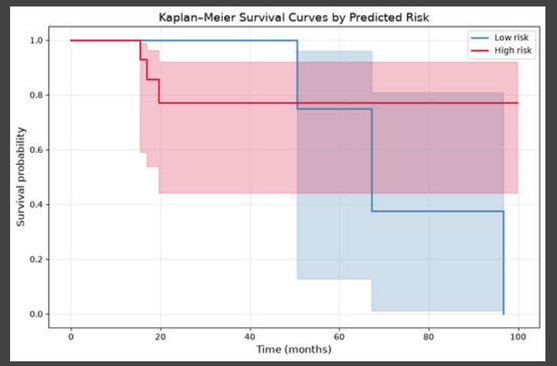
GIGA



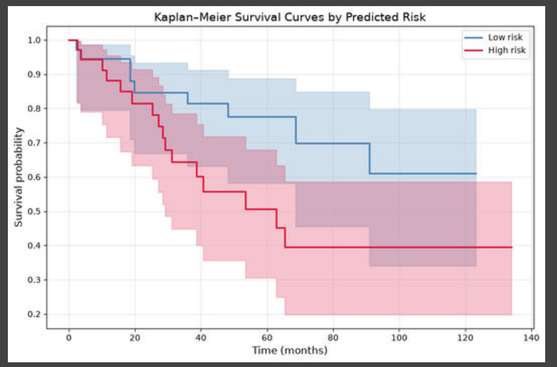
LUAD



LUSC

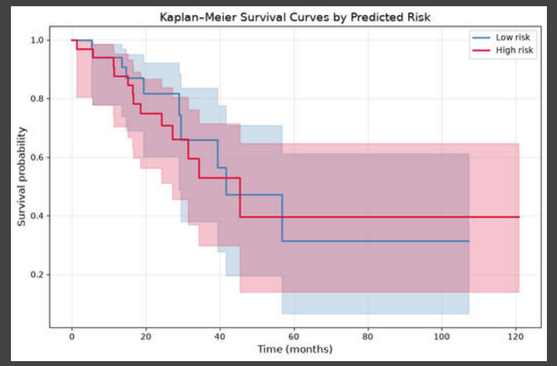


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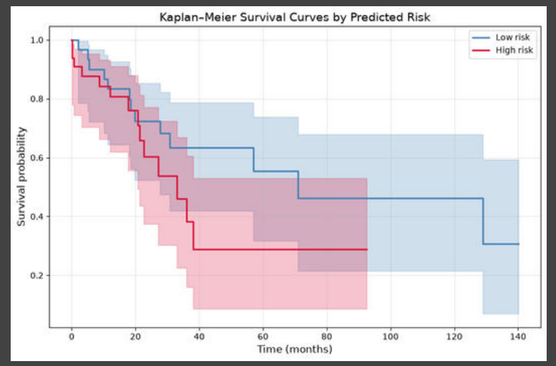


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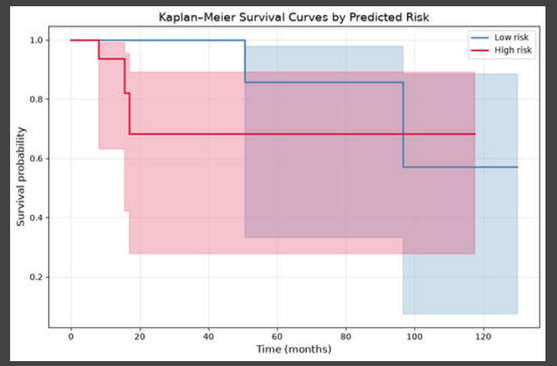
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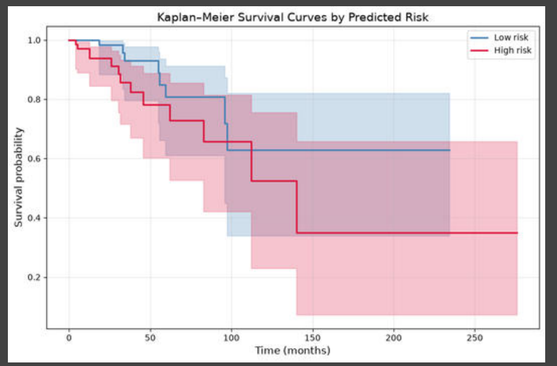
LUAD



LUSC



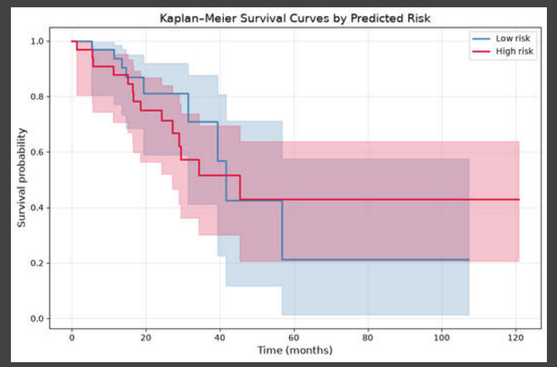
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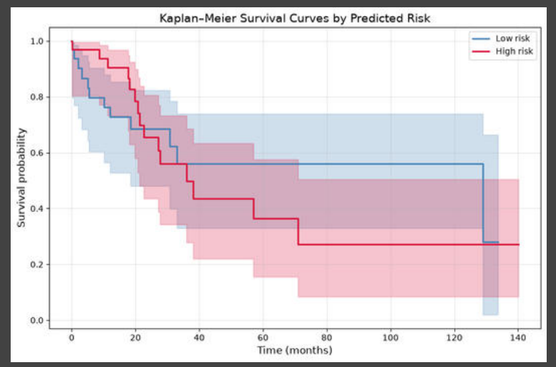
BRCA

CLAM KSMC

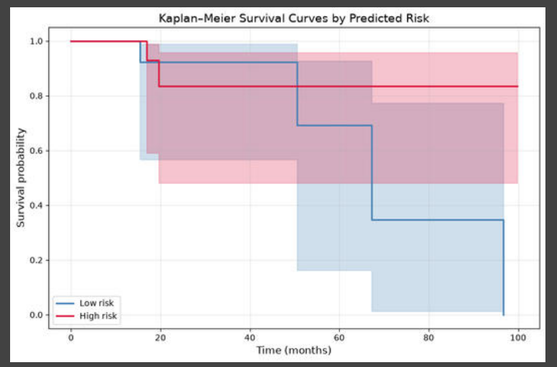
GIGA



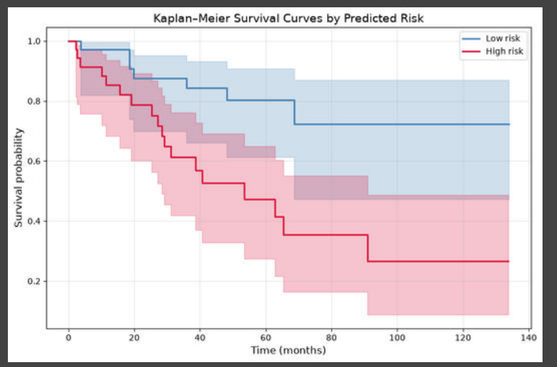
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LUSC

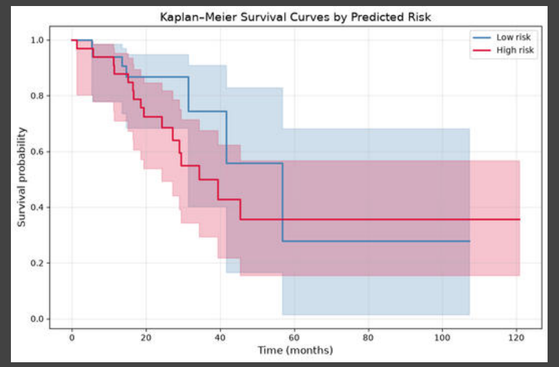


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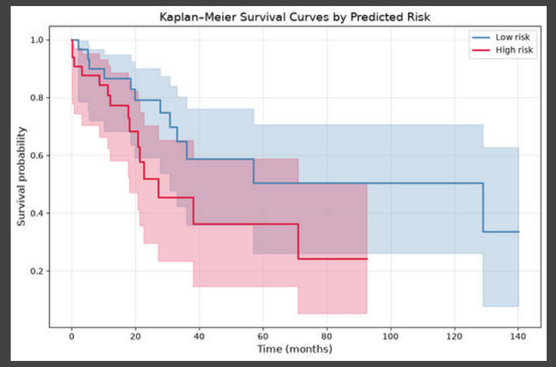


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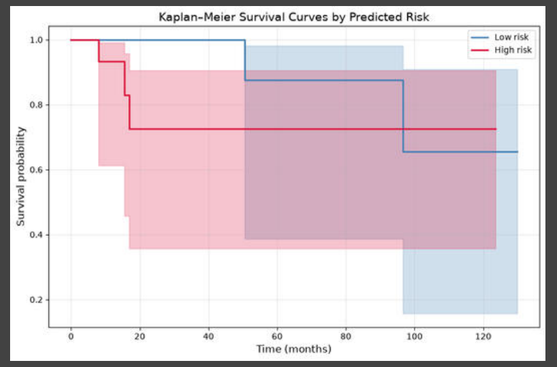
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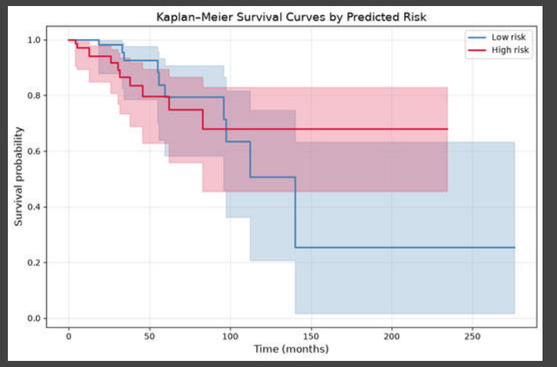
LUAD



LUSC



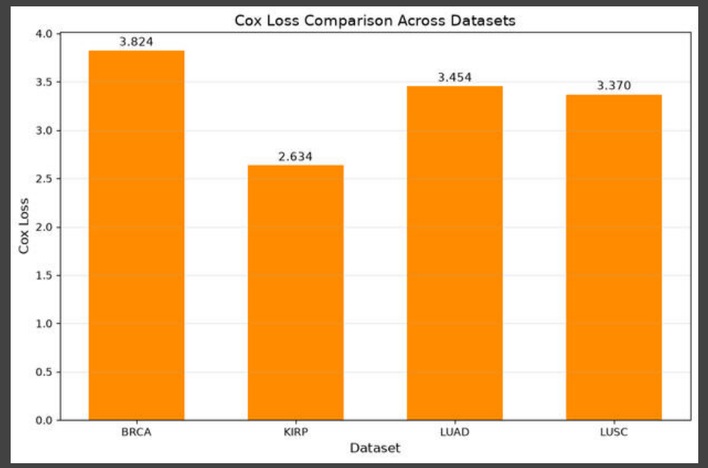
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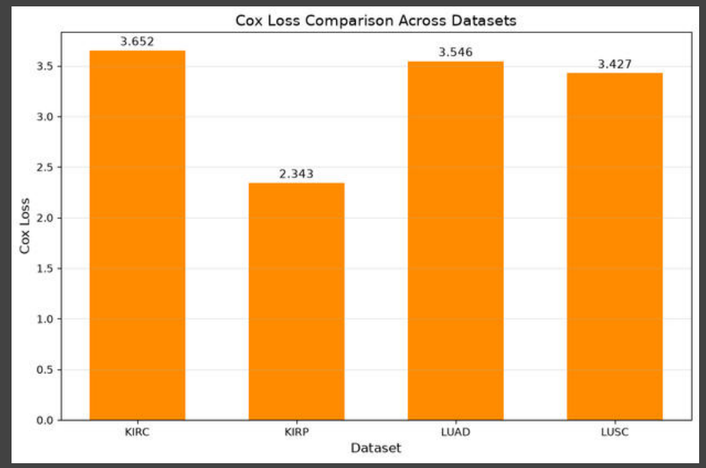
BRCA

Cox Loss

ABMIL

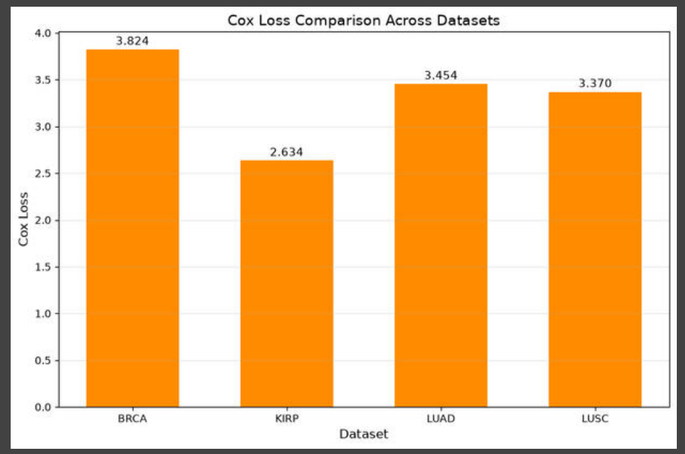


UNI

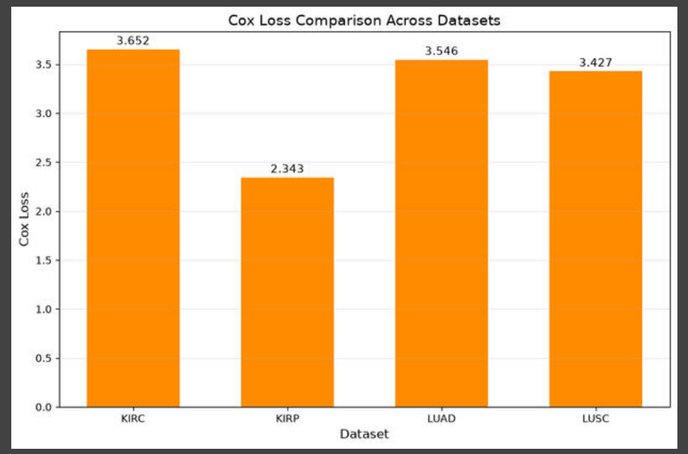


GIGA

CLAM



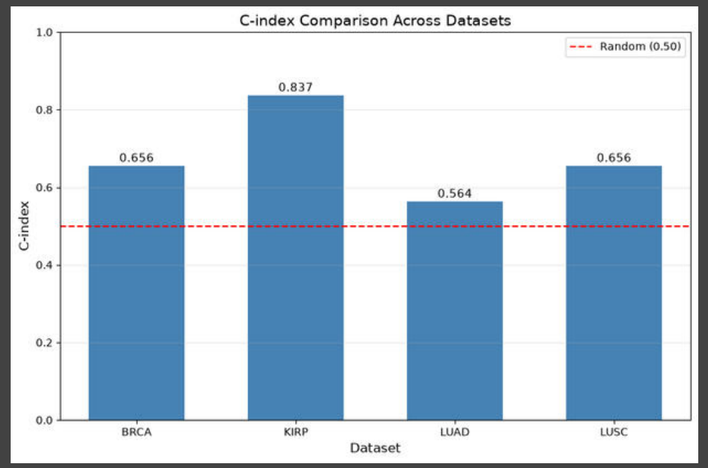
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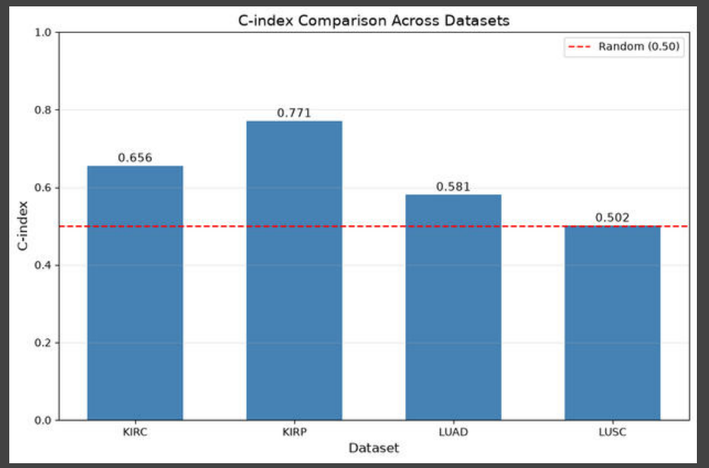
GIGA

C-Index

ABMIL

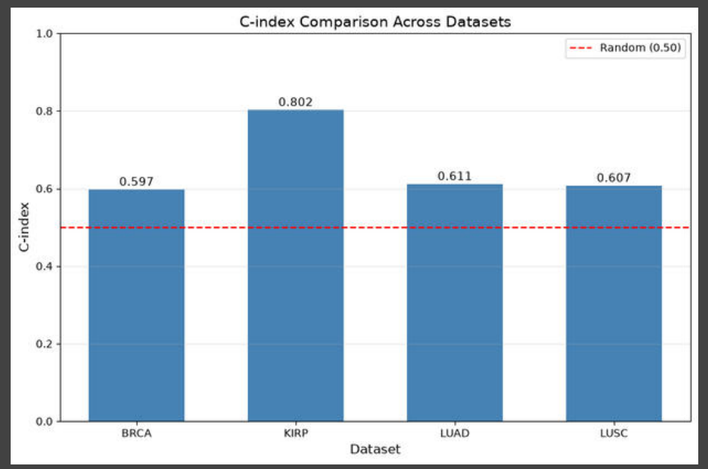


UNI

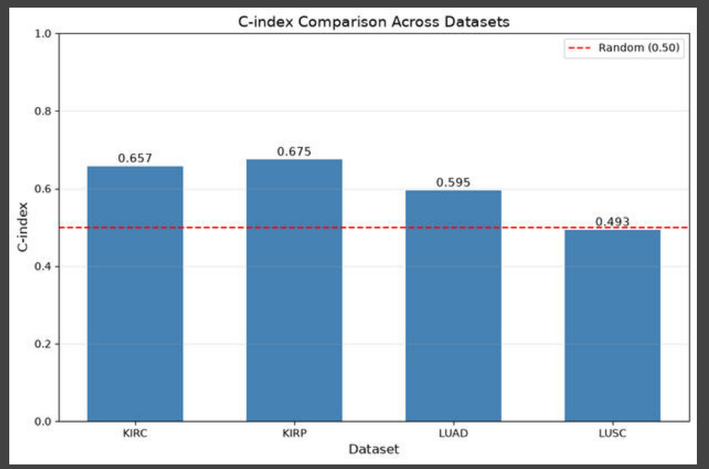


GIGA

CLAM



UNI



GIGA

Conclusion

- UNI and GIGA both provide useful feature representations
- GIGA showed better clustering separation overall
- Clusters contain mixed but meaningful cell-type information
- UNI and GIGA achieved comparable C-index performance
- Survival performance is not directly determined by clustering quality
- Feature representation quality is task-dependent

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Thank You

for your time
and attention

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